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Open Geospatial Consortium

OGC INDOORGML 2.0 PART 2A – XML ENCODING

STANDARD
Implementation

CANDIDATE SWG DRAFT

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CONTENTS

I ABSTRACT

The IndoorGML 2.0 Part 2a – XML Encoding Standard presents the implementation-dependent Geography Markup Language (GML) XML schema encoding of the concepts defined by the IndoorGML 2.0 Part 1 – Conceptual Model Standard (OGC 22-045r5). The XML Encoding is compliant with GML version 3.2.1 as specified by OGC 07-036. The XML encoding of IndoorGML 2.0 defines a concrete implementation schema for representing indoor spatial information, including primal space, dual space, thematic layers, interlayer connections, and navigation-related indoor features. This encoding can be used to store and exchange 2D and/or 3D indoor space and indoor navigation network models in XML format as data sets or via web services.

This Part 2a of the Standard defines how the concepts developed in Part 1 are realized using XML Schema definitions. The XML encoding represents a full mapping of the Conceptual Model and is derived from the UML model following the UML-to-GML/XML encoding rules. Table 1 maps requirements classes from the IndoorGML 2.0 Conceptual Model into the implementation details documented in this standard.

Table 1 — Requirements classes mapping

CONCEPTUAL MODEL	SECTION	XML SCHEMA
IndoorGML Core	Clause 7	Annex B.1
Navigation Extension	Clause 8	Annex B.2

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

OGCDoc, OGC documents, indoor, navigation, IndoorGML, GML, XML

III PREFACE

In order to achieve consensus on the modeling indoor spaces and their neighborhood relationships to support indoor location-based services, a UML Conceptual Model, IndoorGML 2.0 Part 1, was approved as an OGC Standard (OGC 22-045r5) in June, 2025. This model covers geometric and semantic properties of indoor spaces relevant for indoor navigation. This IndoorGML 2.0 Part 2a: XML Encoding Standard defines how those concepts should be realized using XML schema format. The XML encoding is intended for software developers, data providers, standards implementers, and service providers who need a structured, standard, and machine-readable representation of IndoorGML datasets using XML/GML.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- The University of New South Wales
- Pusan National University
- CityGeometrix
- National Institute of Advanced Industrial Science and Technology
- SpatialAlgebra

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SCOPE

1 SCOPE

The OGC® IndoorGML 2.0 Part 2a – XML Encoding Standard defines an XML encoding for IndoorGML 2.0 Part 1 — Conceptual Model. The encoding is implemented as an application schema of the Geography Markup Language version 3.2.1 (GML 3.2.1).

The IndoorGML XML Encoding Standard specifies:

- XML schema for IndoorGML 2.0 core module;
- XML schema for IndoorGML 2.0 navigation module;
- the geometry encoding rules based on GML 3.2.1 geometry objects; and
- a statement to which IndoorGML XML conformance classes an IndoorGML 2.0 instance conforms to.

IndoorGML XML encoding is an implementation encoding of IndoorGML 2.0 Part 1. Therefore, IndoorGML XML instances shall be interpreted according to the semantics defined in the IndoorGML 2.0 conceptual model.

2

CONFORMANCE

2 CONFORMANCE

This standard defines implementation specification which specifies how the IndoorGML 2.0 Conceptual Model should be implemented using Extensible Markup Language (XML) and GML 3.2.1. The Standardization Target for this standard is:

- Implementations of the IndoorGML 2.0 Conceptual Model using XML encodings.

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface standard, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).

All requirements-classes and conformance-classes described in this document are owned by the standard(s) identified.

Table 2 — Conformance classes

CONFORMANCE CLASS	URI
IndoorGML Core	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core
Navigation Extension	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Data elements and interchange formats – Information interchange – Representation of dates and times

Geographic information – Reference model – Part 1: Fundamentals

Geographic Information – Conceptual Schema Language

Geographic information – Conformance and testing

Geographic Information – Spatial Schema

Geographic Information – Rules for Application Schemas

Geographic information – Spatial referencing by coordinates

Geographic Information – Metadata – Part 1: Fundamentals

Geographic Information – Metadata – XML schema implementation

Carl Reed, Open Geospatial Consortium: OGC 04-084, *Topic 0 — Overview*. Open Geospatial Consortium (2005). https://docs.ogc.org/as/04-084/04-084/pdf.

Cliff Kottman, Carl Reed, Open Geospatial Consortium: OGC 08-126, *Topic 5 — Features*. Open Geospatial Consortium (2009). https://docs.ogc.org/as/08-126/08-126/pdf.

Cliff Kottman, Open Geospatial Consortium: OGC 99-108r2, *Topic 8 — Relationships Between Features*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-108r2/99-108r2/pdf.

Cliff Kottman, Open Geospatial Consortium: OGC 99-110, *Topic 10 — Feature Collections*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-110/99-110/pdf.

Clemens Portele, Open Geospatial Consortium: OGC 07-036, *OpenGIS Geography Markup Language (GML) Encoding Standard*. Open Geospatial Consortium (2007). https://docs.ogc.org/is/07-036/07-036/pdf.

Sisi Zlatanova, Abdoulaye Diakite, Taehoon Kim, Ki-Joune Li, Open Geospatial Consortium: OGC 22-045r5, *OGC IndoorGML 2.0 Part 1 – Conceptual Model*. Open Geospatial Consortium (2025). http://www.opengis.net/doc/IS/indoorgml/2.0.

4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. conceptual model

model that defines concepts of a universe of discourse

[**SOURCE:** ISO 19101-1:2014]

4.2. conceptual schema

formal description of a conceptual model

[**SOURCE:** ISO 19101-1:2014]

4.3. feature

abstraction of real world phenomena

Note 1 to entry: Features are objects that have an identity that allows for distinguishing features from each other. Features can have spatial and non-spatial properties that describe the features in more detail. Spatial properties are properties that have a geometry from ISO 19107:2003 as data type.

Note 2 to entry: Features are denoted by the stereotype «FeatureType» in the IndoorGML 2.0 Conceptual Model.

[**SOURCE:** ISO 19101-1:2014]

5

CONVENTIONS

5 CONVENTIONS

5.1. Identifiers

The normative provisions in this standard are denoted by the URI

`http://www.opengis.net/spec/indoorgml-2/2.0/xml`

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Use of XML terminology

The following terms and concepts are used as specified in the W3C XML specifications and OGC GML 3.2.1:

- “element” refers to an XML element tag;
- “attribute” refers to a name/value attribute on an XML element;
- “complexType” refers to an XML Schema complex type definition;
- “substitutionGroup” refers to the GML element substitution mechanism allowing specialized elements to substitute base GML feature types; and
- “property element” refers to an XML element that represents an association between features or objects.

6

OVERVIEW OF THE XML ENCODING

6 OVERVIEW OF THE XML ENCODING

The XML encoding of IndoorGML 2.0 implements the conceptual classes of IndoorGML 2.0 Part 1 as GML-compliant XML elements and types. The conceptual model remains normative for semantics, while this document defines the XML schema representation.

6.1. Design Principle

The XML encoding is based on the following encoding principles.

6.1.1. Conceptual model preservation

The XML encoding of the IndoorGML core module preserves the IndoorGML 2.0 core conceptual model structure, as shown in Figure 1.

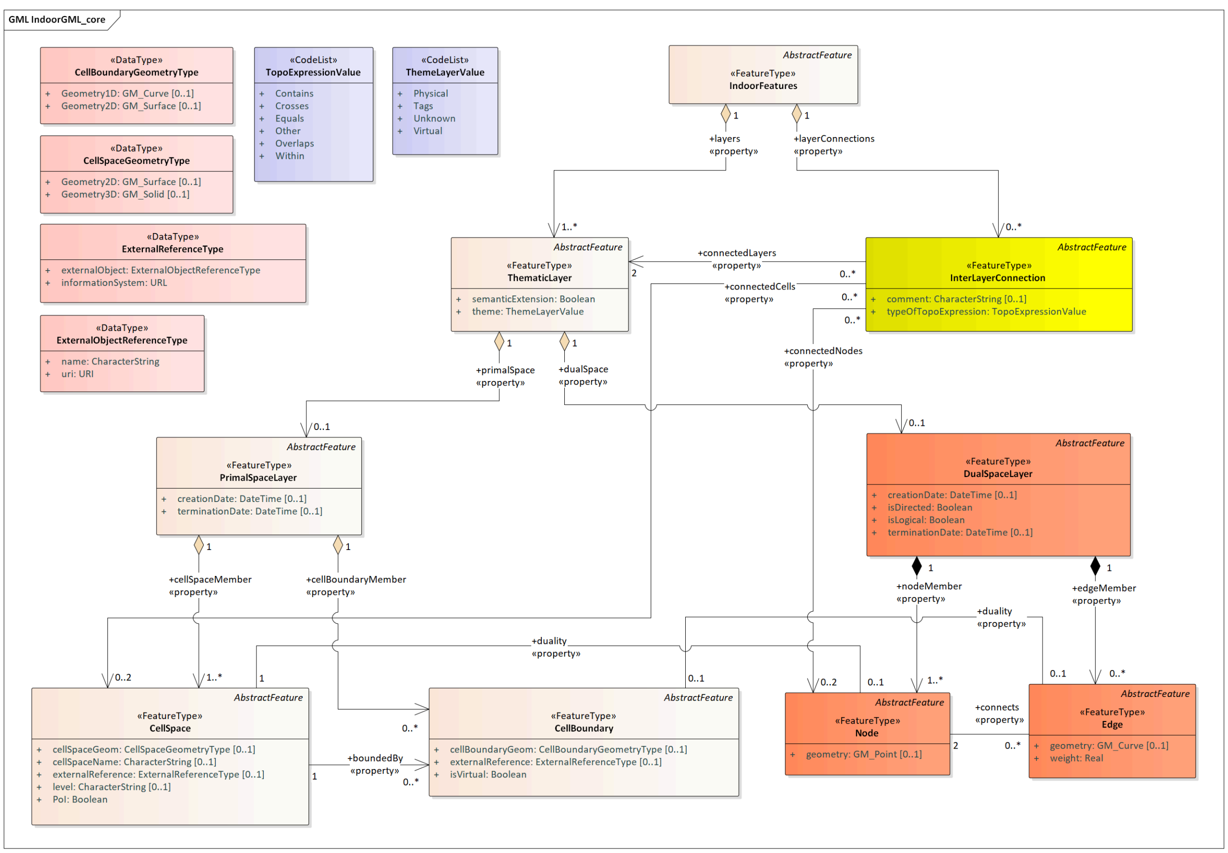

Figure 1 — IndoorGML 2.0 core module structure

Each feature class is mapped to an XML element inheriting from or substituting GML feature elements (e.g. `gml:AbstractFeature`). For example, the `<CellSpace>` element substitutes `gml:AbstractFeature` and extends `gml:AbstractFeatureType` to represent cell spaces.

The XML encoding of the IndoorGML navigation module preserves the IndoorGML 2.0 navigation conceptual model structure, as shown in Figure 2.

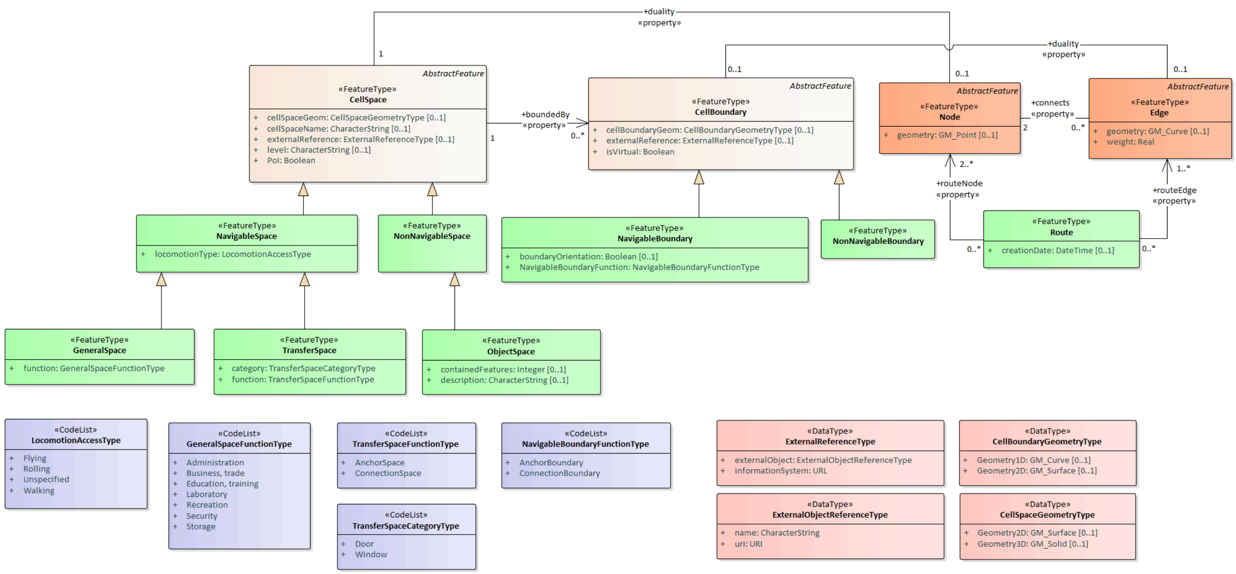

Figure 2 — IndoorGML 2.0 navigation module structure

Navigation feature classes extend core elements using GML inheritance patterns, implementing substitutions where appropriate. For example, `<NavigableSpace>` substitutes `<CellSpace>` and extends its complex type definition.

6.1.2. XML Schema design workflow

The XML encoding schema is derived from the IndoorGML 2.0 UML model and is designed on the basis of OGC GML 3.2.1. XML schema (`.xsd`) documents are generated from the UML model, establishing elements and complex types for the Core and Navigation modules. Typical patterns include converting associations into GML property types using `xlink:href` references (for example, the `duality` link from `CellSpace` to `Node`), mapping attributes to native XML schema elements (such as `creationDate` of type `xs:dateTime`), and representing UML code lists as XML simple types restricting `xs:string` with enumeration constraints.

7 INDOORXML CORE

7 INDOORXML CORE

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
TARGET TYPE	Implementation Specification
CONFORMANCE CLASS	Conformance class A.1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core`
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req http://www.opengis.net/spec/gml/3.2.1/req
NORMATIVE STATEMENTS	Requirement 1: `/req/core/global` Requirement 2: `/req/core/indoorfeatures` Requirement 3: `/req/core/thematiclayer` Requirement 4: `/req/core/primalspacelayer` Requirement 5: `/req/core/cellspace` Requirement 6: `/req/core/cellboundary` Requirement 7: `/req/core/dualspacelayer` Requirement 8: `/req/core/node` Requirement 9: `/req/core/edge` Requirement 10: `/req/core/interlayerconnection`

7.1. General requirements for encoding standard

The IndoorGML Core module defines the foundational features and data structures for representing indoor spaces. It is composed of three main parts:

- The primal space, describing cellular space subdivisions and their boundaries;
- The dual space, carrying network graph information (nodes and edges); and
- The inter-layer connections, describing topological relationships between thematic layers.

An instance XML document representing an IndoorGML 2.0 Core dataset SHALL be well-formed XML and conform to the XML schemas defined in this standard and OGC GML 3.2.1.

REQUIREMENT 1: REQUIREMENT FOR GLOBAL XML SCHEMA VALIDATION	
IDENTIFIER	`/req/core/global`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	An instance XML document representing an IndoorGML 2.0 Core dataset SHALL be well-formed XML and validate against the schemas defined in this standard (including `IndoorGML_core.xsd` and OGC GML 3.2.1).

7.2. IndoorFeatures

The **`IndoorFeatures`** element is an instance of IndoorFeatures, representing the root container feature for an indoor dataset. It aggregates one or more thematic layers and may optionally include interlayer connections.

The **`IndoorFeatures`** element shall contain:

- `gml:id` attribute
- `layers` property containing or referencing **`ThematicLayer`** elements
- optionally, `layerConnections` property containing or referencing **`InterLayerConnection`** elements

```
<element name="IndoorFeatures" substitutionGroup="gml:AbstractFeature" type=
"core:IndoorFeaturesType"/>
<complexType name="IndoorFeaturesType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element maxOccurs="unbounded" name="layers">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:ThematicLayer"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
        <element maxOccurs="unbounded" minOccurs="0" name="layerConnections">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:InterLayerConnection"/>
                </sequence>
```

```xml
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="IndoorFeaturesPropertyType">
  <sequence minOccurs="0">
    <element ref="core:IndoorFeatures"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the IndoorFeatures element

```xml
<core:IndoorFeatures gml:id="IF_1"
    xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
    xmlns:gml="http://www.opengis.net/gml/3.2"
    xmlns:xlink="http://www.w3.org/1999/xlink">
    <!-- mandatory properties: at least one ThematicLayer -->
    <core:layers>
        <core:ThematicLayer gml:id="TL_Pedestrian">...</core:ThematicLayer>
    </core:layers>
    <core:layers>
        <core:ThematicLayer gml:id="TL_Wheelchair">...</core:ThematicLayer>
    </core:layers>

    <!-- optional properties: InterLayerConnection -->
    <core:layerConnections>
        <core:InterLayerConnection gml:id="ILC_1">...</core:InterLayerConnection>
    </core:layerConnections>
</core:IndoorFeatures>
```

Listing — An example of the IndoorFeatures element

REQUIREMENT 2: REQUIREMENT FOR INDOORFEATURES XML ELEMENT	
IDENTIFIER	`/req/core/indoorfeatures`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	An **`IndoorFeatures`** element SHALL contain at least one `layers` property containing or referencing a **`ThematicLayer`** element.
B	When present, each `layerConnections` property of an **`IndoorFeatures`** element SHALL contain or reference an **`InterLayerConnection`** element.
C	An **`IndoorFeatures`** element SHALL contain a unique `gml:id` attribute.

7.3. ThematicLayer

The **ThematicLayer** element is an instance of ThematicLayer, which represents one contextual decomposition of indoor space (e.g. physical layer, navigation layer, sensor layer).

The **ThematicLayer** element shall contain:

- `gml:id` attribute
- `semanticExtension` element indicating whether there is an extension module with additional semantic information
- `theme` element determining what type of model representation can be expected
- optionally, `primalSpace` property for the **PrimalSpaceLayer** element
- optionally, `dualSpace` property for the **DualSpaceLayer** element

```
<element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type="core:
ThematicLayerType"/>
<complexType name="ThematicLayerType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element name="semanticExtension" type="boolean"/>
        <element name="theme" type="core:ThemeLayerValueType"/>
        <element minOccurs="0" name="primalSpace">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:PrimalSpaceLayer"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
        <element minOccurs="0" name="dualSpace">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:DualSpaceLayer"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="ThematicLayerPropertyType">
  <sequence minOccurs="0">
```

```xml
    <element ref="core:ThematicLayer"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the ThematicLayer element

```xml
<core:ThematicLayer gml:id="TL_1">
    <!-- mandatory properties -->
    <core:semanticExtension>true</core:semanticExtension>
    <core:theme>Physical</core:theme>

    <!-- optional properties -->
    <core:primalSpace xlink:href="#PSL_1"/>
    <core:dualSpace xlink:href="#DSL_1"/>
</core:ThematicLayer>
```

Listing — An example of the ThematicLayer element

REQUIREMENT 3: REQUIREMENT FOR THEMATICLAYER XML ELEMENT	
IDENTIFIER	`/req/core/thematiclayer`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	A **`ThematicLayer`** element SHALL declare whether there is an extension module with additional semantic information using the `semanticExtension` element (`true` or `false`).
B	A **`ThematicLayer`** element SHALL determine what type of model representation is expected using the `theme` element.
C	The `theme` value SHALL be one of the following: `"Tags"`, `"Virtual"`, `"Unknown"`, `"Physical"`, or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: WiFi"` or `"other: Sensor"`).
D	When present, the `primalSpace` property of a **`ThematicLayer`** element SHALL contain or reference (via `xlink:href`) a **`PrimalSpaceLayer`** element.
E	When present, the `dualSpace` property of a **`ThematicLayer`** element SHALL contain or reference (via `xlink:href`) a **`DualSpaceLayer`** element.
F	A **`ThematicLayer`** element SHALL contain a unique `gml:id` attribute.

7.4. PrimalSpaceLayer

The **`PrimalSpaceLayer`** element is an instance of PrimalSpaceLayer, which contains spatial objects representing indoor subdivisions in primal space.

The **PrimalSpaceLayer** element shall contain:

- `gml:id` attribute
- `cellSpaceMember` property for representing **CellSpace** elements
- optionally, `cellBoundaryMember` property for representing **CellBoundary** elements
- optionally, `creationDate` element
- optionally, `terminationDate` element

```
<element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type=
"core:PrimalSpaceLayerType"/>
<complexType name="PrimalSpaceLayerType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="creationDate" type="dateTime"/>
        <element minOccurs="0" name="terminationDate" type="dateTime"/>
        <element maxOccurs="unbounded" name="cellSpaceMember">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:CellSpace"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
        <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:CellBoundary"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
      </sequence>
    </extension>
  </complexContent>
</complexType>
```

Listing — XML schema for the PrimalSpaceLayer element

```
<core:PrimalSpaceLayer gml:id="PSL_1">
    <!-- mandatory properties (at least one CellSpace) -->
    <core:cellSpaceMember>
        <core:CellSpace gml:id="CS_1">...</core:CellSpace>
    </core:cellSpaceMember>

    <!-- optional properties -->
    <core:cellBoundaryMember>
        <core:CellBoundary gml:id="CB_1">...</core:CellBoundary>
    </core:cellBoundaryMember>
```

```
    <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:PrimalSpaceLayer>
```

Listing — An example of the PrimalSpaceLayer element

REQUIREMENT 4: REQUIREMENT FOR PRIMALSPACELAYER XML ELEMENT	
IDENTIFIER	`/req/core/primalspacelayer`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	A **`PrimalSpaceLayer`** element SHALL contain at least one `cellSpaceMember` property containing or referencing a **`CellSpace`** element.
B	The **`CellSpace`** elements belonging to the same **`PrimalSpaceLayer`** SHALL not intersect with each other.
C	The **`CellBoundary`** elements belonging to the same **`PrimalSpaceLayer`** SHALL not intersect with each other.
D	When present, `creationDate` and `terminationDate` SHALL be formatted as valid `xs:dateTime` strings.
E	A **`PrimalSpaceLayer`** element SHALL contain a unique `gml:id` attribute.

7.5. CellSpace

The **`CellSpace`** element is an instance of CellSpace, which represents a unit of indoor space such as a room, corridor, stairwell, elevator cabin, or zone.

The **`CellSpace`** element shall contain:

- `gml:id` attribute
- `PoI` element specifying whether this **`CellSpace`** is a point of interest
- optionally, `cellSpaceName` element for the name assigned to the space
- optionally, `level` element
- optionally, `duality` property mapped to the corresponding **`Node`** element in the dual space
- optionally, `cellSpaceGeom` element specifying the 2D or 3D geometry of the cell space
- optionally, `boundedBy` property referencing the linked **`CellBoundary`** elements
- optionally, `externalReference` property for representing **`ExternalReference`** elements

```xml
<element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="core:
CellSpaceType"/>
<complexType name="CellSpaceType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="cellSpaceGeom" type="core:
CellSpaceGeometryPropertyType"/>
        <element minOccurs="0" name="externalReference" type="core:
ExternalReferencePropertyType"/>
        <element minOccurs="0" name="level" type="string"/>
        <element name="PoI" type="boolean"/>
        <element minOccurs="0" name="cellSpaceName" type="string"/>
        <element minOccurs="0" name="duality" type="core:NodePropertyType"/>
        <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="core:
CellBoundaryPropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="CellSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="core:CellSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type=
"core:CellSpaceGeometryType"/>
<complexType name="CellSpaceGeometryType">
  <sequence>
    <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
  </sequence>
</complexType>
<complexType name="CellSpaceGeometryPropertyType">
  <sequence>
    <element ref="core:CellSpaceGeometry"/>
  </sequence>
</complexType>
```

Listing — XML schema for the CellSpace element

```xml
<core:CellSpace gml:id="CS_1">
    <!-- mandatory property -->
    <core:PoI>false</core:PoI>

    <!-- optional properties -->
    <core:cellSpaceGeom>
        <core:Geometry3D>
            <gml:Solid gml:id="CSG_CS_1">
                ...
            </gml:Solid>
        </core:Geometry3D>
    </core:cellSpaceGeom>
    <core:duality xlink:href="#ND_1"/>
    <core:level>3rd Floor</core:level>
    <core:cellSpaceName>Locker Room Lv3</core:cellSpaceName>
    <core:boundedBy xlink:href="#CB_1"/>
</core:CellSpace>
```

Listing — An example of the CellSpace element

REQUIREMENT 5: REQUIREMENT FOR CELLSPACE XML ELEMENT	
IDENTIFIER	`/req/core/cellspace`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	A **`CellSpace`** element SHALL declare whether it is a point of interest space using the `PoI` element (`true` or `false`).
B	When present, the `duality` property of a **`CellSpace`** element SHALL reference (via `xlink:href`) or contain the corresponding **`Node`** element in the dual space.
C	When present, each `boundedBy` property of a **`CellSpace`** element SHALL reference (via `xlink:href`) or contain a corresponding **`CellBoundary`** element.
D	The geometry of each linked **`CellBoundary`** element SHALL not exceed the extent of the **`CellSpace`** geometry (`cellSpaceGeom`).
E	A **`CellSpace`** element SHALL contain a unique `gml:id` attribute.

7.6. CellBoundary

The **`CellBoundary`** element is an instance of CellBoundary, representing a physical or virtual boundary of a cell space.

The **`CellBoundary`** element shall contain:

- `gml:id` attribute
- `isVirtual` element indicating whether a **`CellBoundary`** corresponds to a virtual surface (`true`) or a physical one (`false`)
- optionally, `duality` property mapped to the corresponding **`Edge`** element
- optionally, `cellBoundaryGeom` element specifying the 1D or 2D geometry of the cell boundary
- optionally, `externalReference` property for representing **`ExternalReference`** elements

```
<element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="core:
CellBoundaryType"/>
<complexType name="CellBoundaryType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="cellBoundaryGeom" type="core:CellBoundaryGeo
metryPropertyType"/>
```

```xml
        <element minOccurs="0" name="externalReference" type="core:
ExternalReferencePropertyType"/>
        <element name="isVirtual" type="boolean"/>
        <element minOccurs="0" name="duality" type="core:EdgePropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="CellBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="core:CellBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type=
"core:CellBoundaryGeometryType"/>
<complexType name="CellBoundaryGeometryType">
  <sequence>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
  </sequence>
</complexType>
<complexType name="CellBoundaryGeometryPropertyType">
  <sequence>
    <element ref="core:CellBoundaryGeometry"/>
  </sequence>
</complexType>
```

Listing — XML schema for the CellBoundary element

```xml
<core:CellBoundary gml:id="CB_1">
    <!-- mandatory property -->
    <core:isVirtual>false</core:isVirtual>

    <!-- optional properties -->
    <core:cellBoundaryGeom>
        <core:Geometry2D>
            <gml:Surface gml:id="CBG_CB_1">
                ...
            </gml:Surface>
        </core:Geometry2D>
    </core:cellBoundaryGeom>
    <core:duality xlink:href="#EDG_1"/>
</core:CellBoundary>
```

Listing — An example of the CellBoundary element

REQUIREMENT 6: REQUIREMENT FOR CELLBOUNDARY XML ELEMENT	
IDENTIFIER	`/req/core/cellboundary`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	A **`CellBoundary`** element SHALL specify whether it represents a virtual boundary (`true`) or a physical boundary (`false`) using the `isVirtual` element.

REQUIREMENT 6: REQUIREMENT FOR CELLBOUNDARY XML ELEMENT	
B	When present, the `duality` property of a **`CellBoundary`** element SHALL reference (via `xlink:href`) or contain the corresponding **`Edge`** element in the dual space.
C	A **`CellBoundary`** element SHALL contain a unique `gml:id` attribute.

7.7. DualSpaceLayer

The **`DualSpaceLayer`** element is an instance of DualSpaceLayer, representing the graph structure (nodes and edges) corresponding to a primal space layer.

The **`DualSpaceLayer`** element shall contain:

- `gml:id` attribute
- `isLogical` element indicating whether the graph is logical (`true`) or geometric (`false`)
- `isDirected` element specifying whether the graph is directed
- `nodeMember` property for representing **`Node`** elements
- optionally, `edgeMember` property for representing **`Edge`** elements
- optionally, `creationDate` element
- optionally, `terminationDate` element

```
<element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type=
"core:DualSpaceLayerType"/>
<complexType name="DualSpaceLayerType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element name="isLogical" type="boolean"/>
        <element minOccurs="0" name="creationDate" type="dateTime"/>
        <element minOccurs="0" name="terminationDate" type="dateTime"/>
        <element name="isDirected" type="boolean"/>
        <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
          <complexType>
            <complexContent>
              <extension base="gml:AbstractFeatureMemberType">
                <sequence minOccurs="0">
                  <element ref="core:Edge"/>
                </sequence>
                <attributeGroup ref="gml:AssociationAttributeGroup"/>
              </extension>
            </complexContent>
          </complexType>
        </element>
        <element maxOccurs="unbounded" name="nodeMember">
```

```xml
            <complexType>
              <complexContent>
                <extension base="gml:AbstractFeatureMemberType">
                  <sequence minOccurs="0">
                    <element ref="core:Node"/>
                  </sequence>
                  <attributeGroup ref="gml:AssociationAttributeGroup"/>
                </extension>
              </complexContent>
            </complexType>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="DualSpaceLayerPropertyType">
    <sequence minOccurs="0">
      <element ref="core:DualSpaceLayer"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
```

Listing — XML schema for the DualSpaceLayer element

```xml
<core:DualSpaceLayer gml:id="DSL_1">
    <!-- mandatory properties -->
    <core:isLogical>false</core:isLogical>
    <core:isDirected>false</core:isDirected>
    <core:nodeMember>
        <core:Node gml:id="ND_1">...</core:Node>
    </core:nodeMember>

    <!-- optional properties -->
    <core:edgeMember>
        <core:Edge gml:id="EDG_1">...</core:Edge>
    </core:edgeMember>
    <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:DualSpaceLayer>
```

Listing — An example of the DualSpaceLayer element

REQUIREMENT 7: REQUIREMENT FOR DUALSPACELAYER XML ELEMENT	
IDENTIFIER	`/req/core/dualspacelayer`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	A **`DualSpaceLayer`** element SHALL specify whether the graph is logical (`true`) or geometric (`false`) using the `isLogical` element.
B	When `isLogical` is `false`, the geometries of all **`Node`** instances contained in `nodeMember` SHALL be spatially located inside their corresponding **`CellSpace`** geometries.
C	A **`DualSpaceLayer`** element SHALL specify whether the graph is directed using the `isDirected` element (`true` or `false`).

REQUIREMENT 7: REQUIREMENT FOR DUALSPACELAYER XML ELEMENT	
D	A **DualSpaceLayer** element SHALL contain at least one `nodeMember` property containing or referencing a **Node** element.
E	A **DualSpaceLayer** element SHALL contain a unique `gml:id` attribute.

7.8. Node

The **Node** element is an instance of Node, representing a state or graph vertex in the dual space.

The **Node** element shall contain:

- `gml:id` attribute
- `duality` property referencing the corresponding **CellSpace** element
- optionally, `geometry` element representing the position of the node
- optionally, `connects` property referencing connected **Edge** elements

```
<element name="Node" substitutionGroup="gml:AbstractFeature" type="core:
NodeType"/>
<complexType name="NodeType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
        <element name="duality" type="core:CellSpacePropertyType"/>
        <element maxOccurs="unbounded" minOccurs="0" name="connects" type="core:
EdgePropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="NodePropertyType">
  <sequence minOccurs="0">
    <element ref="core:Node"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the Node element

```
<core:Node gml:id="ND_1">
    <!-- mandatory property -->
    <core:duality xlink:href="#CS_1"/>

    <!-- optional properties -->
    <core:geometry>
        <gml:Point gml:id="G_ND_1">
```

```
            <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
        </gml:Point>
    </core:geometry>
    <core:connects xlink:href="#EDG_1"/>
</core:Node>
```

Listing — An example of the Node element

REQUIREMENT 8: REQUIREMENT FOR NODE XML ELEMENT	
IDENTIFIER	/req/core/node
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A **Node** element SHALL contain a duality property referencing (via xlink:href) or containing the corresponding **CellSpace** element in the primal space.
B	When present, each connects property of a **Node** element SHALL reference (via xlink:href) or contain a connected **Edge** element.
C	A **Node** element SHALL contain a unique gml:id attribute.

7.9. Edge

The **Edge** element is an instance of Edge, which represents an adjacency, connectivity, or navigability relationship between two nodes in dual space.

The **Edge** element shall contain:

- gml:id attribute
- weight element which can be used in graph-based applications
- connects property referencing exactly two connected **Node** elements
- optionally, duality property referencing the corresponding **CellBoundary** element
- optionally, geometry element representing the curve of the edge

```
<element name="Edge" substitutionGroup="gml:AbstractFeature" type="core:
EdgeType"/>
<complexType name="EdgeType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
        <element name="weight" type="double"/>
```

```xml
        <element minOccurs="0" name="duality" type="core:
CellBoundaryPropertyType"/>
        <element maxOccurs="2" minOccurs="2" name="connects" type="core:
NodePropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="EdgePropertyType">
  <sequence minOccurs="0">
    <element ref="core:Edge"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the Edge element

```xml
<core:Edge gml:id="EDG_1">
    <!-- mandatory properties -->
    <core:weight>1.0</core:weight>
    <core:connects xlink:href="#ND_1"/>
    <core:connects xlink:href="#ND_2"/>

    <!-- optional properties -->
    <core:geometry>
        <gml:LineString gml:id="G_EDG_1">
            <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
            <gml:pos>445537.914644321 5444904.59001251 -2.02</gml:pos>
        </gml:LineString>
    </core:geometry>
    <core:duality xlink:href="#CB_1"/>
</core:Edge>
```

Listing — An example of the Edge element

REQUIREMENT 9: REQUIREMENT FOR EDGE XML ELEMENT	
IDENTIFIER	`/req/core/edge`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	An **Edge** element SHALL contain exactly two `connects` properties referencing (via `xlink:href`) or containing the connected **Node** elements.
B	If `isDirected` is `true` in the parent **DualSpaceLayer**, the sequence of `connects` properties SHALL define the direction of the edge (from the first node to the second node).
C	An **Edge** element SHALL contain a `weight` element of type `xs:double`.
D	When present, the curve in `geometry` SHALL not self-intersect, and its start and end coordinates SHALL match the coordinates of the connected **Node** instances referenced by `connects`.
E	When present, the `duality` property of an **Edge** element SHALL reference (via `xlink:href`) or contain the corresponding **CellBoundary** element.

REQUIREMENT 9: REQUIREMENT FOR EDGE XML ELEMENT	
F	An **Edge** element SHALL contain a unique `gml:id` attribute.

7.10. InterLayerConnection

The **`InterLayerConnection`** element is an instance of InterLayerConnection, which describes a topological relationship between objects in different thematic layers.

The **`InterLayerConnection`** element shall contain:

- `gml:id` attribute
- `connectedLayers` property mapped to exactly two **`ThematicLayer`** elements
- `typeOfTopoExpression` element specifying the topological relationship
- optionally, `connectedNodes` property mapped to two **`Node`** elements
- optionally, `connectedCells` property mapped to two **`CellSpace`** elements
- optionally, `comment` element

The `typeOfTopoExpression` element represents the topological relationship between connected elements across different thematic layers, with values including `"Contains"`, `"Overlaps"`, `"Equals"`, `"Within"`, `"Crosses"`, and `"Other"`. These topological values are conceptualized as verbs in a Subject-Verb-Object (SVO) sentence structure:

- When `connectedCells` (or `connectedNodes`) are specified, the topological relationship primarily applies to those specific cell spaces (or nodes). That is, the first referenced cell or node acts as the Subject, and the second referenced cell or node acts as the Object (e.g., *Cell A of Layer 1 Contains Cell B of Layer 2*, even if Layer 1 as a whole does not fully contain Layer 2).
- In the absence of `connectedCells` and `connectedNodes`, the topological relationship applies directly between the two whole thematic layers, where the first referenced layer in `connectedLayers` is the Subject and the second is the Object (e.g., *Layer 1 Contains Layer 2*).

Consequently, the sequence of elements across `connectedLayers`, `connectedCells`, and `connectedNodes` properties shall strictly maintain this pairwise SVO order, where index 1 always represents the Subject (or the element from the Subject layer) and index 2 represents the Object (or the element from the Object layer).

```
<element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature"
type="core:InterLayerConnectionType"/>
```

```xml
<complexType name="InterLayerConnectionType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element name="typeOfTopoExpression" type="core:TopoExpressionValueType"/
>
        <element minOccurs="0" name="comment" type="string"/>
        <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="core:
ThematicLayerPropertyType"/>
        <element maxOccurs="2" minOccurs="0" name="connectedCells" type="core:
CellSpacePropertyType"/>
        <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="core:
NodePropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="InterLayerConnectionPropertyType">
  <sequence minOccurs="0">
    <element ref="core:InterLayerConnection"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="TopoExpressionValueType">
  <union memberTypes="core:TopoExpressionValueEnumerationType core:
TopoExpressionValueOtherType"/>
</simpleType>
<simpleType name="TopoExpressionValueEnumerationType">
  <restriction base="string">
    <enumeration value="Contains"/>
    <enumeration value="Overlaps"/>
    <enumeration value="Equals"/>
    <enumeration value="Within"/>
    <enumeration value="Crosses"/>
    <enumeration value="Other"/>
  </restriction>
</simpleType>
<simpleType name="TopoExpressionValueOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the InterLayerConnection element

```xml
<core:InterLayerConnection gml:id="ILC_1">
    <!-- mandatory properties -->
    <core:typeOfTopoExpression>Contains</core:typeOfTopoExpression>
    <core:connectedLayers xlink:href="#TL_Pedestrian"/>
    <core:connectedLayers xlink:href="#TL_Wheelchair"/>

    <!-- optional properties -->
    <core:comment>Pedestrian layer contains wheelchair layer subdivision</core:
comment>
    <core:connectedCells xlink:href="#CS_Ped_1"/>
    <core:connectedCells xlink:href="#CS_Wheel_1"/>
</core:InterLayerConnection>
```

Listing — An example of the InterLayerConnection element

REQUIREMENT 10: REQUIREMENT FOR INTERLAYERCONNECTION XML ELEMENT	
IDENTIFIER	`/req/core/interlayerconnection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
A	An **`InterLayerConnection`** element SHALL contain exactly two `connectedLayers` properties referencing (via `xlink:href`) or containing two distinct **`ThematicLayer`** elements.
B	An **`InterLayerConnection`** element SHALL specify the topological relationship using the `typeOfTopoExpression` element, which applies primarily to the referenced `connectedCells` (or `connectedNodes`), or to the `connectedLayers` when no cells or nodes are referenced.
C	The `typeOfTopoExpression` value SHALL be one of the following: `"Contains"`, `"Overlaps"`, `"Equals"`, `"Within"`, `"Crosses"`, `"Other"`, or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: Touches"`).
D	When present, `connectedCells` SHALL contain exactly two properties referencing (via `xlink:href`) or containing **`CellSpace`** elements belonging to the primal space layers of the first and second connected thematic layers, respectively.
E	When present, `connectedNodes` SHALL contain exactly two properties referencing (via `xlink:href`) or containing **`Node`** elements belonging to the dual space layers of the first and second connected thematic layers, respectively.
F	The sequence of elements in the `connectedLayers`, `connectedCells`, and `connectedNodes` properties SHALL be ordered such that the first element always represents the subject of the topological relationship, and the second element represents the object, ensuring they pairwise correspond at the same indices.
G	An **`InterLayerConnection`** element SHALL contain a unique `gml:id` attribute.

7.11. ExternalReference

The **`ExternalReference`** element is used for referencing an object in an external dataset from an IndoorGML feature.

```
<element name="ExternalReference" substitutionGroup="gml:AbstractObject" type=
"core:ExternalReferenceType"/>
<complexType name="ExternalReferenceType">
  <sequence>
    <element name="externalObject" type="core:ExternalObjectReferencePropertyType
"/>
    <element name="informationSystem" type="gmd:URL_PropertyType"/>
  </sequence>
</complexType>
<element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject"
type="core:ExternalObjectReferenceType"/>
<complexType name="ExternalObjectReferenceType">
```

```xml
  <sequence>
    <element name="name" type="string"/>
    <element name="uri" type="anyURI"/>
  </sequence>
</complexType>
```

Listing — XML schema for the ExternalReference element

```xml
<core:externalReference>
    <core:ExternalReference>
        <core:externalObject>
            <core:ExternalObjectReference>
                <core:name>IFC_Space_101</core:name>
                <core:uri>urn:uuid:2b88523c-a9e3-4c92-9442-5f6966f3630f</core:
uri>
            </core:ExternalObjectReference>
        </core:externalObject>
        <core:informationSystem>
            <gmd:URL>https://bim.example.org/project/buildingA.ifc</gmd:URL>
        </core:informationSystem>
    </core:ExternalReference>
</core:externalReference>
```

Listing — An example of the ExternalReference element

8

INDOORXML NAVIGATION

8 INDOORXML NAVIGATION

REQUIREMENTS CLASS 2	
IDENTIFIER	`http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
TARGET TYPE	Implementation Specification
CONFORMANCE CLASS	Conformance class A.2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi`
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
NORMATIVE STATEMENTS	Requirement 11: `/req/navi/global` Requirement 12: `/req/navi/navigablespace` Requirement 13: `/req/navi/generalspace` Requirement 14: `/req/navi/transferspace` Requirement 15: `/req/navi/navigableboundary` Requirement 16: `/req/navi/nonnavigablespace` Requirement 17: `/req/navi/objectspace` Requirement 18: `/req/navi/nonnavigableboundary` Requirement 19: `/req/navi/route`

8.1. General requirements for encoding standards

The IndoorGML Navigation extension module provides semantic information and specialized feature types to support indoor navigation applications. It extends the Core module by classifying indoor spaces and boundaries into navigable and non-navigable categories, and by providing representations for designated navigation routes.

An instance XML document representing an IndoorGML 2.0 Navigation module dataset SHALL be well-formed XML and conform to the XML schemas defined in this standard (`IndoorGML_navi.xsd`, `IndoorGML_core.xsd`) and OGC GML 3.2.1.

REQUIREMENT 11: REQUIREMENT FOR NAVIGATION MODULE XML SCHEMA VALIDATION	
IDENTIFIER	`/req/navi/global`

REQUIREMENT 11: REQUIREMENT FOR NAVIGATION MODULE XML SCHEMA VALIDATION	
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	An instance XML document representing an IndoorGML 2.0 Navigation module dataset SHALL be well-formed XML and validate against the schemas defined in this standard (including `IndoorGML_navi.xsd`, `IndoorGML_core.xsd`, and OGC GML 3.2.1).

8.2. NavigableSpace

The **`NavigableSpace`** element is an instance of NavigableSpace, representing a space where an agent or user can move freely (e.g. corridor, door, lobby, hallway, room).

The **`NavigableSpace`** element shall contain:

- all properties inherited from the **`CellSpace`** element
- `locomotionType` element indicating the allowed locomotion mode in indoor space

```
<element name="NavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NavigableSpaceType"/>
<complexType name="NavigableSpaceType">
  <complexContent>
    <extension base="core:CellSpaceType">
      <sequence>
        <element name="locomotionType" type="navi:LocomotionAccessType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="NavigableSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:NavigableSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="LocomotionAccessType">
  <union memberTypes="navi:LocomotionAccessEnumerationType navi:
LocomotionAccessOtherType"/>
</simpleType>
<simpleType name="LocomotionAccessEnumerationType">
  <restriction base="string">
    <enumeration value="Walking"/>
    <enumeration value="Flying"/>
    <enumeration value="Rolling"/>
    <enumeration value="Unspecified"/>
  </restriction>
</simpleType>
<simpleType name="LocomotionAccessOtherType">
  <restriction base="string">
```

```
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the NavigableSpace element

```
<navi:NavigableSpace gml:id="NS_1">
    <!-- inherited mandatory property from CellSpace -->
    <core:PoI>false</core:PoI>

    <!-- mandatory property -->
    <navi:locomotionType>Walking</navi:locomotionType>
</navi:NavigableSpace>
```

Listing — An example of the NavigableSpace element

REQUIREMENT 12: REQUIREMENT FOR NAVIGABLESPACE XML ELEMENT	
IDENTIFIER	`/req/navi/navigablespace`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`NavigableSpace`** element SHALL conform to all schema and semantic requirements of **`CellSpace`**.
B	A **`NavigableSpace`** element SHALL specify the allowed locomotion mode using the `locomotionType` element.
C	The `locomotionType` value SHALL be one of the following: `"Walking"`, `"Flying"`, `"Rolling"`, `"Unspecified"`, or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: Drone"` or `"other: Segway"`).
D	A **`NavigableSpace`** element SHALL contain a unique `gml:id` attribute.

8.3. GeneralSpace

The **`GeneralSpace`** element is an instance of GeneralSpace, representing a navigable space that agents can occupy for extended periods and which can serve as starting or destination points in navigation.

The **`GeneralSpace`** element shall contain:

- all properties inherited from the **`NavigableSpace`** element
- `function` element specifying details about the functional purpose of the space

```
<element name="GeneralSpace" substitutionGroup="navi:NavigableSpace" type="navi:
GeneralSpaceType"/>
```

```xml
<complexType name="GeneralSpaceType">
  <complexContent>
    <extension base="navi:NavigableSpaceType">
      <sequence>
        <element name="function" type="navi:GeneralSpaceFunctionType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="GeneralSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:GeneralSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="GeneralSpaceFunctionType">
  <union memberTypes="navi:GeneralSpaceFunctionEnumerationType navi:
GeneralSpaceFunctionOtherType"/>
</simpleType>
<simpleType name="GeneralSpaceFunctionEnumerationType">
  <restriction base="string">
    <enumeration value="Administration"/>
    <enumeration value="Business, trade"/>
    <enumeration value="Education, training"/>
    <enumeration value="Recreation"/>
    <enumeration value="Laboratory"/>
    <enumeration value="Storage"/>
    <enumeration value="Security"/>
  </restriction>
</simpleType>
<simpleType name="GeneralSpaceFunctionOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the GeneralSpace element

```xml
<navi:GeneralSpace gml:id="GS_1">
    <!-- inherited mandatory properties -->
    <core:PoI>false</core:PoI>
    <navi:locomotionType>Walking</navi:locomotionType>

    <!-- mandatory property -->
    <navi:function>Storage</navi:function>
</navi:GeneralSpace>
```

Listing — An example of the GeneralSpace element

REQUIREMENT 13: REQUIREMENT FOR GENERALSPACE XML ELEMENT	
IDENTIFIER	`/req/navi/generalspace`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`GeneralSpace`** element SHALL conform to all schema and semantic requirements of **`NavigableSpace`**.

REQUIREMENT 13: REQUIREMENT FOR GENERALSPACE XML ELEMENT	
B	A **GeneralSpace** element SHALL specify the functional purpose of the space using the `function` element.
C	The `function` value SHALL be one of the following: `"Administration"`, `"Business, trade"`, `"Education, training"`, `"Recreation"`, `"Laboratory"`, `"Storage"`, `"Security"`, or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: Cafeteria"`).
D	A **GeneralSpace** element SHALL contain a unique `gml:id` attribute.

8.4. TransferSpace

The **TransferSpace** element is an instance of TransferSpace, representing a navigable space that provides passage between general spaces (e.g. doors, windows, gates).

The **TransferSpace** element shall contain:

- all properties inherited from the **NavigableSpace** element
- `function` element describing whether the space is an "AnchorSpace" or a "ConnectionSpace"
- `type` element specifying opening categories (e.g., Door or Window)

Note that an AnchorSpace is a space that allows for the connection between the indoor and the outdoor. A ConnectionSpace is a space connecting two indoor or two outdoor spaces.

```
<element name="TransferSpace" substitutionGroup="navi:NavigableSpace" type="navi:
TransferSpaceType"/>
<complexType name="TransferSpaceType">
  <complexContent>
    <extension base="navi:NavigableSpaceType">
      <sequence>
        <element name="function" type="navi:TransferSpaceFunctionType"/>
        <element name="type" type="navi:TransferSpaceTypeType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="TransferSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:TransferSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="TransferSpaceFunctionType">
```

```xml
  <union memberTypes="navi:TransferSpaceFunctionEnumerationType navi:TransferSpac
eFunctionOtherType"/>
</simpleType>
<simpleType name="TransferSpaceFunctionEnumerationType">
  <restriction base="string">
    <enumeration value="ConnectionSpace"/>
    <enumeration value="AnchorSpace"/>
  </restriction>
</simpleType>
<simpleType name="TransferSpaceFunctionOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
<simpleType name="TransferSpaceTypeType">
  <union memberTypes="navi:TransferSpaceTypeEnumerationType navi:
TransferSpaceTypeOtherType"/>
</simpleType>
<simpleType name="TransferSpaceTypeEnumerationType">
  <restriction base="string">
    <enumeration value="Door"/>
    <enumeration value="Window"/>
  </restriction>
</simpleType>
<simpleType name="TransferSpaceTypeOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the TransferSpace element

```xml
<navi:TransferSpace gml:id="TS_1">
    <!-- inherited mandatory properties -->
    <core:PoI>false</core:PoI>
    <navi:locomotionType>Walking</navi:locomotionType>

    <!-- mandatory properties -->
    <navi:function>ConnectionSpace</navi:function>
    <navi:type>Door</navi:type>
</navi:TransferSpace>
```

Listing — An example of the TransferSpace element

REQUIREMENT 14: REQUIREMENT FOR TRANSFERSPACE XML ELEMENT	
IDENTIFIER	`/req/navi/transferspace`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`TransferSpace`** element SHALL conform to all schema and semantic requirements of **`NavigableSpace`**.
B	A **`TransferSpace`** element SHALL specify its passage function using the `function` element and its opening category using the `type` element.
C	The `function` value SHALL be one of the following: `"ConnectionSpace"` (connecting two indoor or two outdoor spaces), `"AnchorSpace"` (enabling connection between indoor and

REQUIREMENT 14: REQUIREMENT FOR TRANSFERSPACE XML ELEMENT	
	outdoor spaces), or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: InterbuildingSpace"`).
D	The `type` value SHALL be one of the following: `"Door"`, `"Window"`, or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: Turnstile"` or `"other: Elevator"`).
E	A **`TransferSpace`** element SHALL contain a unique `gml:id` attribute.

8.5. NavigableBoundary

The **`NavigableBoundary`** element is an instance of NavigableBoundary, representing a boundary of a navigable space that provides transition or connection information.

The **`NavigableBoundary`** element shall contain:

- all properties inherited from the **`CellBoundary`** element
- `NavigableBoundaryFunction` element describing whether the boundary is an "AnchorBoundary" or a "ConnectionBoundary"
- optionally, `boundaryOrientation` element indicating whether to use the geometry represented by `cellBoundaryGeom` as-is (`true`) or in reverse (`false`)

Note that an AnchorBoundary is the boundary of an AnchorSpace that enables the connection between indoor and outdoor spaces. A ConnectionBoundary is the boundary of a ConnectionSpace that connects two indoor spaces or two outdoor spaces.

```
<element name="NavigableBoundary" substitutionGroup="core:CellBoundary" type=
"navi:NavigableBoundaryType"/>
<complexType name="NavigableBoundaryType">
  <complexContent>
    <extension base="core:CellBoundaryType">
      <sequence>
        <element name="NavigableBoundaryFunction" type="navi:
NavigableBoundaryFunctionType"/>
        <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="NavigableBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="navi:NavigableBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="NavigableBoundaryFunctionType">
```

```xml
  <union memberTypes="navi:NavigableBoundaryFunctionEnumerationType navi:Navigabl
eBoundaryFunctionOtherType"/>
</simpleType>
<simpleType name="NavigableBoundaryFunctionEnumerationType">
  <restriction base="string">
    <enumeration value="AncorBoundary"/>
    <enumeration value="ConnectionBoundary"/>
  </restriction>
</simpleType>
<simpleType name="NavigableBoundaryFunctionOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the NavigableBoundary element

```xml
<navi:NavigableBoundary gml:id="NB_1">
    <!-- inherited mandatory property from CellBoundary -->
    <core:isVirtual>false</core:isVirtual>

    <!-- mandatory property -->
    <navi:NavigableBoundaryFunction>ConnectionBoundary</navi:
NavigableBoundaryFunction>

    <!-- optional property -->
    <navi:boundaryOrientation>true</navi:boundaryOrientation>
</navi:NavigableBoundary>
```

Listing — An example of the NavigableBoundary element

REQUIREMENT 15: REQUIREMENT FOR NAVIGABLEBOUNDARY XML ELEMENT	
IDENTIFIER	`/req/navi/navigableboundary`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`NavigableBoundary`** element SHALL conform to all schema and semantic requirements of **`CellBoundary`**.
B	A **`NavigableBoundary`** element SHALL specify its boundary function using the `NavigableBoundaryFunction` element.
C	The `NavigableBoundaryFunction` value SHALL be one of the following: `"ConnectionBoundary"` (boundary connecting two indoor or two outdoor spaces), `"AncorBoundary"` (boundary enabling transition between indoor and outdoor spaces), or a user-defined value matching the pattern `"other: \w{2,}"` (e.g., `"other: VirtualBoundary"`).
D	When present, `boundaryOrientation` SHALL specify whether the boundary geometry is used with standard orientation (`true`) or reversed orientation (`false`).
E	A **`NavigableBoundary`** element SHALL contain a unique `gml:id` attribute.

8.6. NonNavigableSpace

The **NonNavigableSpace** element is an instance of NonNavigableSpace, representing an indoor space subdivision occupied by obstacles or restricted from agent traversal.

The **NonNavigableSpace** element shall contain:

- all properties inherited from the **CellSpace** element

```
<element name="NonNavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NonNavigableSpaceType"/>
<complexType name="NonNavigableSpaceType">
  <complexContent>
    <extension base="core:CellSpaceType">
      <sequence/>
    </extension>
  </complexContent>
</complexType>
<complexType name="NonNavigableSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:NonNavigableSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the NonNavigableSpace element

```
<navi:NonNavigableSpace gml:id="NNS_1">
    <!-- inherited mandatory property from CellSpace -->
    <core:PoI>false</core:PoI>
</navi:NonNavigableSpace>
```

Listing — An example of the NonNavigableSpace element

REQUIREMENT 16: REQUIREMENT FOR NONNAVIGABLESPACE XML ELEMENT	
IDENTIFIER	`/req/navi/nonnavigablespace`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **NonNavigableSpace** element SHALL conform to all schema and semantic requirements of **CellSpace**.
B	A **NonNavigableSpace** element SHALL contain a unique `gml:id` attribute.

8.7. ObjectSpace

The **ObjectSpace** element is an instance of ObjectSpace, representing a specialized non-navigable space that encapsulates physical objects or furniture within an indoor environment.

The **ObjectSpace** element shall contain:

- all properties inherited from the **NonNavigableSpace** element
- optionally, `description` element providing descriptive text regarding the contained objects
- optionally, `containedFeatures` element representing the number of objects encapsulated within the **ObjectSpace**

```
<element name="ObjectSpace" substitutionGroup="navi:NonNavigableSpace" type=
"navi:ObjectSpaceType"/>
<complexType name="ObjectSpaceType">
  <complexContent>
    <extension base="navi:NonNavigableSpaceType">
      <sequence>
        <element minOccurs="0" name="description" type="string"/>
        <element minOccurs="0" name="containedFeatures" type="integer"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="ObjectSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:ObjectSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the ObjectSpace element

```
<navi:ObjectSpace gml:id="OS_1">
    <!-- inherited mandatory property from CellSpace -->
    <core:PoI>false</core:PoI>

    <!-- optional properties -->
    <navi:containedFeatures>3</navi:containedFeatures>
    <navi:description>Living room furniture and columns</navi:description>
</navi:ObjectSpace>
```

Listing — An example of the ObjectSpace element

REQUIREMENT 17: REQUIREMENT FOR OBJECTSPACE XML ELEMENT	
IDENTIFIER	`/req/navi/objectspace`

REQUIREMENT 17: REQUIREMENT FOR OBJECTSPACE XML ELEMENT	
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	An **`ObjectSpace`** element SHALL conform to all schema and semantic requirements of **`NonNavigableSpace`**.
B	An **`ObjectSpace`** element SHALL NOT overlap with any other **`CellSpace`** or its specialized classes within the same **`PrimalSpaceLayer`**.
C	When present, `containedFeatures` SHALL be a non-negative integer representing the count of contained objects.
D	When present, `description` SHALL provide descriptive text regarding the contained objects.
E	An **`ObjectSpace`** element SHALL contain a unique `gml:id` attribute.

8.8. NonNavigableBoundary

The **`NonNavigableBoundary`** element is an instance of NonNavigableBoundary, representing the boundary between two non-navigable spaces or between a navigable space and a non-navigable space.

The **`NonNavigableBoundary`** element shall contain:

- all properties inherited from the **`CellBoundary`** element

```
<element name="NonNavigableBoundary" substitutionGroup="core:CellBoundary" type=
"navi:NonNavigableBoundaryType"/>
<complexType name="NonNavigableBoundaryType">
  <complexContent>
    <extension base="core:CellBoundaryType">
      <sequence/>
    </extension>
  </complexContent>
</complexType>
<complexType name="NonNavigableBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="navi:NonNavigableBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the NonNavigableBoundary element

```
<navi:NonNavigableBoundary gml:id="NNB_1">
    <!-- inherited mandatory property from CellBoundary -->
    <core:isVirtual>false</core:isVirtual>
```

```
</navi:NonNavigableBoundary>
```

Listing — An example of the NonNavigableBoundary element

REQUIREMENT 18: REQUIREMENT FOR NONNAVIGABLEBOUNDARY XML ELEMENT	
IDENTIFIER	`/req/navi/nonnavigableboundary`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`NonNavigableBoundary`** element SHALL conform to all schema and semantic requirements of **`CellBoundary`**.
B	A **`NonNavigableBoundary`** element SHALL contain a unique `gml:id` attribute.

8.9. Route

The **`Route`** element is an instance of Route, representing a designated navigation path through an indoor space composed of an ordered sequence of nodes and connecting edges in dual space.

The **`Route`** element shall contain:

- `gml:id` attribute
- `routeNode` properties containing ordered sequences of **`Node`** references to describe the different parts of the route path
- `routeEdge` properties containing ordered sequences of **`Edge`** references to describe the different parts of the route path
- optionally, `creationDate` element indicating when a specific route was created

```
<element name="Route" substitutionGroup="gml:AbstractFeature" type="navi:
RouteType"/>
<complexType name="RouteType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="creationDate" type="dateTime"/>
        <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type="core:
NodePropertyType"/>
        <element maxOccurs="unbounded" name="routeEdge" type="core:
EdgePropertyType"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="RoutePropertyType">
```

```xml
  <sequence minOccurs="0">
    <element ref="navi:Route"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the Route element

```xml
<navi:Route gml:id="RT_1">
    <!-- mandatory properties (at least 2 nodes and 1 edge) -->
    <navi:routeNode xlink:href="#ND_1"/>
    <navi:routeNode xlink:href="#ND_2"/>
    <navi:routeEdge xlink:href="#EDG_1"/>

    <!-- optional property -->
    <navi:creationDate>2025-10-20T16:44:00Z</navi:creationDate>
</navi:Route>
```

Listing — An example of the Route element

REQUIREMENT 19: REQUIREMENT FOR ROUTE XML ELEMENT	
IDENTIFIER	`/req/navi/route`
INCLUDED IN	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
A	A **`Route`** element SHALL contain at least two `routeNode` properties and at least one `routeEdge` property.
B	Each `routeNode` property SHALL reference (via `xlink:href`) or contain a **`Node`** element.
C	Each `routeEdge` property SHALL reference (via `xlink:href`) or contain an **`Edge`** element.
D	The sequences of `routeNode` and `routeEdge` properties SHALL represent an ordered contiguous path in the dual space graph.
E	Each consecutive pair of `routeNode` elements in the sequence SHALL be connected by the corresponding `routeEdge` element in `routeEdge`.
F	Consecutive `routeEdge` elements in the sequence SHALL share a common `routeNode` element.
G	When present, `creationDate` SHALL be formatted as a valid `xs:dateTime` string.
H	A **`Route`** element SHALL contain a unique `gml:id` attribute.

A

ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE

A.1. Conformance Class Core

CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core`
REQUIREMENTS CLASS	Requirements class 1: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core`
TARGET TYPE	Implementation Specification
CONFORMANCE TESTS	Abstract test A.1: `/conf/core/global` Abstract test A.2: `/conf/core/indoorfeatures` Abstract test A.3: `/conf/core/thematiclayer` Abstract test A.4: `/conf/core/primalspacelayer` Abstract test A.5: `/conf/core/cellspace` Abstract test A.6: `/conf/core/cellboundary` Abstract test A.7: `/conf/core/dualspacelayer` Abstract test A.8: `/conf/core/node` Abstract test A.9: `/conf/core/edge` Abstract test A.10: `/conf/core/interlayerconnection`

ABSTRACT TEST A.1	
IDENTIFIER	`/conf/core/global`
REQUIREMENT	Requirement 1: `/req/core/global`
TEST PURPOSE	Verify that instance XML documents validate against the IndoorGML 2.0 Core XML schema (`IndoorGML_core.xsd`) and OGC GML 3.2.1 schemas.

ABSTRACT TEST A.1

TEST METHOD	Validate the XML instance document against `IndoorGML_core.xsd` and supporting GML 3.2.1 schemas using an XML Schema validator or appropriate validation software.

ABSTRACT TEST A.2

IDENTIFIER	`/conf/core/indoorfeatures`
REQUIREMENT	Requirement 2: `/req/core/indoorfeatures`
TEST PURPOSE	Verify that an **`IndoorFeatures`** element contains at least one `layers` property referencing or containing a **`ThematicLayer`** element, that optional `layerConnections` properties contain or reference **`InterLayerConnection`** elements, and that a unique `gml:id` is present.
TEST METHOD	Inspect the instance document to verify that the `layers` property has at least one valid **`ThematicLayer`** instance, that `layerConnections` contain valid **`InterLayerConnection`** instances, and that `gml:id` is unique.

ABSTRACT TEST A.3

IDENTIFIER	`/conf/core/thematiclayer`
REQUIREMENT	Requirement 3: `/req/core/thematiclayer`
TEST PURPOSE	Verify that for each **`ThematicLayer`** element: • `semanticExtension` is a valid boolean value (`true` or `false`). • `theme` contains a valid value (`"Tags"`, `"Virtual"`, `"Unknown"`, `"Physical"`, or `"other: \w{2,}"` such as `"other: WiFi"`). • If present, `primalSpace` references or contains a valid **`PrimalSpaceLayer`** element. • If present, `dualSpace` references or contains a valid **`DualSpaceLayer`** element. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify the above properties.

ABSTRACT TEST A.4

IDENTIFIER	`/conf/core/primalspacelayer`
REQUIREMENT	Requirement 4: `/req/core/primalspacelayer`
TEST PURPOSE	Verify that for each **`PrimalSpaceLayer`** element:

ABSTRACT TEST A.4	
	• At least one `cellSpaceMember` property containing or referencing a **`CellSpace`** element is present. • The geometries of all **`CellSpace`** members belonging to the same **`PrimalSpaceLayer`** do not intersect each other. • The geometries of all **`CellBoundary`** members belonging to the same **`PrimalSpaceLayer`** do not intersect each other. • If present, `creationDate` and `terminationDate` are valid `xs:dateTime` strings. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document and perform spatial intersection analysis on member geometries to verify non-overlapping constraints.

ABSTRACT TEST A.5	
IDENTIFIER	`/conf/core/cellspace`
REQUIREMENT	Requirement 5: `/req/core/cellspace`
TEST PURPOSE	Verify that for each **`CellSpace`** element: • `PoI` is a valid boolean value (`true` or `false`). • If present, `duality` references or contains a corresponding **`Node`** element in dual space. • If present, each `boundedBy` references or contains a corresponding **`CellBoundary`** element. • The geometry of each bounding **`CellBoundary`** does not exceed the geometric extent of the **`CellSpace`**. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document and verify geometric boundary containment against the cell space geometry.

ABSTRACT TEST A.6	
IDENTIFIER	`/conf/core/cellboundary`
REQUIREMENT	Requirement 6: `/req/core/cellboundary`
TEST PURPOSE	Verify that for each **`CellBoundary`** element: • `isVirtual` is a valid boolean value (`true` or `false`). • If present, `duality` references or contains a corresponding **`Edge`** element in dual space. • A unique `gml:id` attribute is present.

ABSTRACT TEST A.6	
TEST METHOD	Inspect the instance document to verify the above properties.

ABSTRACT TEST A.7	
IDENTIFIER	`/conf/core/dualspacelayer`
REQUIREMENT	Requirement 7: `/req/core/dualspacelayer`
TEST PURPOSE	Verify that for each **`DualSpaceLayer`** element: • `isLogical` and `isDirected` are valid boolean values (`true` or `false`). • When `isLogical` is `false`, the geometry of each **`Node`** in `nodeMember` is spatially located inside the geometry of its corresponding **`CellSpace`**. • At least one `nodeMember` property containing or referencing a **`Node`** element is present. • If present, `edgeMember` properties contain or reference **`Edge`** elements. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document and perform point-in-polygon / point-in-solid geometric tests when `isLogical` is `false`.

ABSTRACT TEST A.8	
IDENTIFIER	`/conf/core/node`
REQUIREMENT	Requirement 8: `/req/core/node`
TEST PURPOSE	Verify that for each **`Node`** element: • `duality` references or contains the corresponding **`CellSpace`** element in the primal space. • If present, each `connects` property references or contains a connected **`Edge`** element. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify cross-references between nodes, edges, and cell spaces.

ABSTRACT TEST A.9	
IDENTIFIER	`/conf/core/edge`
REQUIREMENT	Requirement 9: `/req/core/edge`

ABSTRACT TEST A.9

TEST PURPOSE	Verify that for each **Edge** element: • Exactly two `connects` properties referencing or containing connected **Node** elements are present. • If the parent **DualSpaceLayer** has `isDirected = true`, the order of `connects` is interpreted as directional (start to end node). • A valid `weight` value of type `xs:double` is present. • If `geometry` is present, the curve does not self-intersect, and its start and end coordinates match the positions of the connected **Node** instances. • If present, `duality` references or contains the corresponding **CellBoundary** element. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document and verify geometric coordinate alignment with connected node positions.

ABSTRACT TEST A.10

IDENTIFIER	`/conf/core/interlayerconnection`
REQUIREMENT	Requirement 10: `/req/core/interlayerconnection`
TEST PURPOSE	Verify that for each **InterLayerConnection** element: • Exactly two `connectedLayers` properties referencing distinct **ThematicLayer** elements are present. • `typeOfTopoExpression` is one of the valid enumeration or pattern values (`"Contains"`, `"Overlaps"`, `"Equals"`, `"Within"`, `"Crosses"`, `"Other"`, or `"other: \w{2,}"` such as `"other: Touches"`). • If present, `connectedCells` contains exactly two references to **CellSpace** elements belonging respectively to the first and second connected thematic layers. • If present, `connectedNodes` contains exactly two references to **Node** elements belonging respectively to the first and second connected thematic layers. • The sequence of `connectedLayers`, `connectedCells`, and `connectedNodes` properties is ordered such that the first element represents the subject of the topological relationship, and the second element represents the object, corresponding pairwise to the same indices. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify layer and object cross-references and topological expression values.

A.2. Conformance Class Navigation Extension

CONFORMANCE CLASS A.2	
IDENTIFIER	`http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi`
REQUIREMENTS CLASS	Requirements class 2: `http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi`
TARGET TYPE	Implementation Specification
CONFORMANCE TESTS	Abstract test A.11: `/conf/navi/global` Abstract test A.12: `/conf/navi/navigablespace` Abstract test A.13: `/conf/navi/generalspace` Abstract test A.14: `/conf/navi/transferspace` Abstract test A.15: `/conf/navi/navigableboundary` Abstract test A.16: `/conf/navi/nonnavigablespace` Abstract test A.17: `/conf/navi/objectspace` Abstract test A.18: `/conf/navi/nonnavigableboundary` Abstract test A.19: `/conf/navi/route`

ABSTRACT TEST A.11	
IDENTIFIER	`/conf/navi/global`
REQUIREMENT	Requirement 11: `/req/navi/global`
TEST PURPOSE	Verify that instance XML documents using the Navigation module validate against `IndoorGML_navi.xsd`, `IndoorGML_core.xsd`, and OGC GML 3.2.1 schemas.
TEST METHOD	Validate the XML instance document against the schemas using an XML Schema validator or appropriate validation software.

ABSTRACT TEST A.12	
IDENTIFIER	`/conf/navi/navigablespace`
REQUIREMENT	Requirement 12: `/req/navi/navigablespace`
TEST PURPOSE	Verify that for each **`NavigableSpace`** element: • All validation criteria for `/conf/core/cellspace` are satisfied.

ABSTRACT TEST A.12	
	• `locomotionType` contains a valid value (`"Walking"`, `"Flying"`, `"Rolling"`, `"Unspecified"`, or `"other: \w{2,}"` such as `"other: Drone"`). • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify locomotionType and inherited CellSpace properties.

ABSTRACT TEST A.13	
IDENTIFIER	`/conf/navi/generalspace`
REQUIREMENT	Requirement 13: `/req/navi/generalspace`
TEST PURPOSE	Verify that for each **`GeneralSpace`** element: • All validation criteria for `/conf/navi/navigablespace` are satisfied. • `function` contains a valid value (`"Administration"`, `"Business, trade"`, `"Education, training"`, `"Recreation"`, `"Laboratory"`, `"Storage"`, `"Security"`, or `"other: \w{2,}"` such as `"other: Cafeteria"`). • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify function type and inherited properties.

ABSTRACT TEST A.14	
IDENTIFIER	`/conf/navi/transferspace`
REQUIREMENT	Requirement 14: `/req/navi/transferspace`
TEST PURPOSE	Verify that for each **`TransferSpace`** element: • All validation criteria for `/conf/navi/navigablespace` are satisfied. • `function` contains a valid value (`"ConnectionSpace"`, `"AnchorSpace"`, or `"other: \w{2,}"` such as `"other: InterbuildingSpace"`). • `type` contains a valid value (`"Door"`, `"Window"`, or `"other: \w{2,}"` such as `"other: Turnstile"`). • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify function, opening type, and inherited properties.

ABSTRACT TEST A.15	
IDENTIFIER	`/conf/navi/navigableboundary`
REQUIREMENT	Requirement 15: `/req/navi/navigableboundary`
TEST PURPOSE	Verify that for each **`NavigableBoundary`** element: • All validation criteria for `/conf/core/cellboundary` are satisfied. • `NavigableBoundaryFunction` contains a valid value (`"ConnectionBoundary"`, `"AncorBoundary"` / `"AnchorBoundary"`, or `"other: \w{2,}"` such as `"other: VirtualBoundary"`). • If present, `boundaryOrientation` is a valid boolean value (`true` or `false`). • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify boundary function and orientation.

ABSTRACT TEST A.16	
IDENTIFIER	`/conf/navi/nonnavigablespace`
REQUIREMENT	Requirement 16: `/req/navi/nonnavigablespace`
TEST PURPOSE	Verify that for each **`NonNavigableSpace`** element: • All validation criteria for `/conf/core/cellspace` are satisfied. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify conformity with CellSpace requirements.

ABSTRACT TEST A.17	
IDENTIFIER	`/conf/navi/objectspace`
REQUIREMENT	Requirement 17: `/req/navi/objectspace`
TEST PURPOSE	Verify that for each **`ObjectSpace`** element: • All validation criteria for `/conf/navi/nonnavigablespace` are satisfied. • The geometry does not overlap with any other **`CellSpace`** or its specialized classes in the same **`PrimalSpaceLayer`**. • If present, `containedFeatures` is a valid non-negative integer. • If present, `description` is a valid string. • A unique `gml:id` attribute is present.

ABSTRACT TEST A.17	
TEST METHOD	Inspect the instance document and verify non-overlapping geometric relationships in the primal space.

ABSTRACT TEST A.18	
IDENTIFIER	`/conf/navi/nonnavigableboundary`
REQUIREMENT	Requirement 18: `/req/navi/nonnavigableboundary`
TEST PURPOSE	Verify that for each **`NonNavigableBoundary`** element: • All validation criteria for `/conf/core/cellboundary` are satisfied. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document to verify conformity with CellBoundary requirements.

ABSTRACT TEST A.19	
IDENTIFIER	`/conf/navi/route`
REQUIREMENT	Requirement 19: `/req/navi/route`
TEST PURPOSE	Verify that for each **`Route`** element: • At least two `routeNode` properties and at least one `routeEdge` property are present. • Each `routeNode` references or contains a **`Node`** element within the same dual space. • Each `routeEdge` references or contains an **`Edge`** element within the same dual space. • `routeNode` and `routeEdge` sequences represent an ordered contiguous path where each consecutive pair of nodes is connected by the corresponding edge, and consecutive edges share a common node. • If present, `creationDate` is a valid `xs:dateTime` string. • A unique `gml:id` attribute is present.
TEST METHOD	Inspect the instance document and perform graph continuity validation along the route path.

B

ANNEX B (NORMATIVE) XML SCHEMA FOR INDOORGML 2

ANNEX B (NORMATIVE) XML SCHEMA FOR INDOORGML 2

In addition to this document, this standard includes some normative XML Schema Documents. These XML Schema Documents are included in this document. These XML Schema Documents are also posted online at the URL http://schemas.opengis.net/indoorgml/2.0. In the event of a discrepancy between this document and online versions of the XML Schema Document files (posted at schemas.opengis.net), the online schema files SHALL be considered authoritative.

B.1. IndoorGML Core Module

```
<?xml version="1.0" encoding="UTF-8"?>
<schema
    xmlns="http://www.w3.org/2001/XMLSchema"
 xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
 xmlns:gmd="http://www.isotc211.org/2005/gmd"
 xmlns:gml="http://www.opengis.net/gml/3.2"
 elementFormDefault="qualified"
 targetNamespace="http://www.opengis.net/indoorgml/2.0/core"
 version="2.0">
  <annotation>
    <appinfo>
      <gml:gmlProfileSchema>IndoorGML</gml:gmlProfileSchema>
    </appinfo>
  </annotation>
  <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://
schemas.opengis.net/iso/19139/20070417/gmd/gmd.xsd"/>
  <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://
schemas.opengis.net/gml/3.2.1/gml.xsd"/>
  <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
  <element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type=
"core:CellBoundaryType"/>
  <complexType name="CellBoundaryType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="cellBoundaryGeom" type="core:CellBoundaryG
eometryPropertyType"/>
          <element minOccurs="0" name="externalReference" type="core:
ExternalReferencePropertyType"/>
          <element name="isVirtual" type="boolean"/>
          <element minOccurs="0" name="duality" type="core:EdgePropertyType">
            <annotation>
              <documentation>Corresponding cell boundary in the primal space</
documentation>
              <appinfo>
                <gml:reversePropertyName>core:duality</gml:reversePropertyName>
              </appinfo>
```

```
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="CellBoundaryPropertyType">
    <sequence minOccurs="0">
      <element ref="core:CellBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject"
type="core:CellBoundaryGeometryType"/>
  <complexType name="CellBoundaryGeometryType">
    <sequence>
      <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
      <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
    </sequence>
  </complexType>
  <complexType name="CellBoundaryGeometryPropertyType">
    <sequence>
      <element ref="core:CellBoundaryGeometry"/>
    </sequence>
  </complexType>
  <element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="core:
CellSpaceType"/>
  <complexType name="CellSpaceType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="cellSpaceGeom" type="core:
CellSpaceGeometryPropertyType"/>
          <element minOccurs="0" name="externalReference" type="core:
ExternalReferencePropertyType"/>
          <element minOccurs="0" name="level" type="string"/>
          <element name="PoI" type="boolean"/>
          <element minOccurs="0" name="cellSpaceName" type="string"/>
          <element minOccurs="0" name="duality" type="core:NodePropertyType">
            <annotation>
              <documentation>Corresponding cell space in the primal space</
documentation>
              <appinfo>
                <gml:reversePropertyName>core:duality</gml:reversePropertyName>
              </appinfo>
            </annotation>
          </element>
          <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type=
"core:CellBoundaryPropertyType">
            <annotation>
              <documentation>Cell boundaries that are bounding a cell space</
documentation>
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="CellSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="core:CellSpace"/>
    </sequence>
```

```xml
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type=
"core:CellSpaceGeometryType"/>
  <complexType name="CellSpaceGeometryType">
    <sequence>
      <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
      <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    </sequence>
  </complexType>
  <complexType name="CellSpaceGeometryPropertyType">
    <sequence>
      <element ref="core:CellSpaceGeometry"/>
    </sequence>
  </complexType>
  <element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type=
"core:DualSpaceLayerType"/>
  <complexType name="DualSpaceLayerType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element name="isLogical" type="boolean"/>
          <element minOccurs="0" name="creationDate" type="dateTime"/>
          <element minOccurs="0" name="terminationDate" type="dateTime"/>
          <element name="isDirected" type="boolean"/>
          <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
            <annotation>
              <documentation>Edges of a dual space</documentation>
            </annotation>
            <complexType>
              <complexContent>
                <extension base="gml:AbstractFeatureMemberType">
                  <sequence minOccurs="0">
                    <element ref="core:Edge"/>
                  </sequence>
                  <attributeGroup ref="gml:AssociationAttributeGroup"/>
                </extension>
              </complexContent>
            </complexType>
          </element>
          <element maxOccurs="unbounded" name="nodeMember">
            <annotation>
              <documentation>Nodes of a dual space</documentation>
            </annotation>
            <complexType>
              <complexContent>
                <extension base="gml:AbstractFeatureMemberType">
                  <sequence minOccurs="0">
                    <element ref="core:Node"/>
                  </sequence>
                  <attributeGroup ref="gml:AssociationAttributeGroup"/>
                </extension>
              </complexContent>
            </complexType>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="DualSpaceLayerPropertyType">
    <sequence minOccurs="0">
      <element ref="core:DualSpaceLayer"/>
```

```xml
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Edge" substitutionGroup="gml:AbstractFeature" type="core:
EdgeType"/>
  <complexType name="EdgeType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
          <element name="weight" type="double"/>
          <element minOccurs="0" name="duality" type="core:
CellBoundaryPropertyType">
            <annotation>
              <documentation>Corresponding edge in the dual space</documentation>
              <appinfo>
                <gml:reversePropertyName>core:duality</gml:reversePropertyName>
              </appinfo>
            </annotation>
          </element>
          <element maxOccurs="2" minOccurs="2" name="connects" type="core:
NodePropertyType">
            <annotation>
              <documentation>The two nodes connected by an edge</documentation>
              <appinfo>
                <gml:reversePropertyName>core:connects</gml:reversePropertyName>
              </appinfo>
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="EdgePropertyType">
    <sequence minOccurs="0">
      <element ref="core:Edge"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject"
type="core:ExternalObjectReferenceType"/>
  <complexType name="ExternalObjectReferenceType">
    <sequence>
      <element name="name" type="string"/>
      <element name="uri" type="anyURI"/>
    </sequence>
  </complexType>
  <complexType name="ExternalObjectReferencePropertyType">
    <sequence>
      <element ref="core:ExternalObjectReference"/>
    </sequence>
  </complexType>
  <element name="ExternalReference" substitutionGroup="gml:AbstractObject" type=
"core:ExternalReferenceType"/>
  <complexType name="ExternalReferenceType">
    <sequence>
      <element name="externalObject" type="core:ExternalObjectReferencePropertyTy
pe"/>
      <element name="informationSystem" type="gmd:URL_PropertyType"/>
    </sequence>
  </complexType>
```

```xml
  <complexType name="ExternalReferencePropertyType">
    <sequence>
      <element ref="core:ExternalReference"/>
    </sequence>
  </complexType>
  <element name="IndoorFeatures" substitutionGroup="gml:AbstractFeature" type=
"core:IndoorFeaturesType"/>
  <complexType name="IndoorFeaturesType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element maxOccurs="unbounded" name="layers">
            <complexType>
              <complexContent>
                <extension base="gml:AbstractFeatureMemberType">
                  <sequence minOccurs="0">
                    <element ref="core:ThematicLayer"/>
                  </sequence>
                  <attributeGroup ref="gml:AssociationAttributeGroup"/>
                </extension>
              </complexContent>
            </complexType>
          </element>
          <element maxOccurs="unbounded" minOccurs="0" name="layerConnections">
            <complexType>
              <complexContent>
                <extension base="gml:AbstractFeatureMemberType">
                  <sequence minOccurs="0">
                    <element ref="core:InterLayerConnection"/>
                  </sequence>
                  <attributeGroup ref="gml:AssociationAttributeGroup"/>
                </extension>
              </complexContent>
            </complexType>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="IndoorFeaturesPropertyType">
    <sequence minOccurs="0">
      <element ref="core:IndoorFeatures"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature"
type="core:InterLayerConnectionType">
    <annotation>
      <documentation>Describes the connection between two thematic layers.</
documentation>
    </annotation>
  </element>
  <complexType name="InterLayerConnectionType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element name="typeOfTopoExpression" type="core:
TopoExpressionValueType">
            <annotation>
              <documentation>Describes the topological relationship between two
layers (e.g. overlaps, contains, etc.).</documentation>
            </annotation>
```

```xml
          </element>
          <element minOccurs="0" name="comment" type="string"/>
          <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="core:
ThematicLayerPropertyType"/>
          <element maxOccurs="2" minOccurs="0" name="connectedCells" type="core:
CellSpacePropertyType">
            <annotation>
              <documentation>Indicates the interconnected primal spaces</
documentation>
            </annotation>
          </element>
          <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="core:
NodePropertyType">
            <annotation>
              <documentation>Indicates the interconnected dual spaces</
documentation>
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="InterLayerConnectionPropertyType">
    <sequence minOccurs="0">
      <element ref="core:InterLayerConnection"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Node" substitutionGroup="gml:AbstractFeature" type="core:
NodeType"/>
  <complexType name="NodeType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
          <element name="duality" type="core:CellSpacePropertyType">
            <annotation>
              <documentation>Corresponding node in the dual space</documentation>
              <appinfo>
                <gml:reversePropertyName>core:duality</gml:reversePropertyName>
              </appinfo>
            </annotation>
          </element>
          <element maxOccurs="unbounded" minOccurs="0" name="connects" type=
"core:EdgePropertyType">
            <annotation>
              <documentation>Edges connected to the node</documentation>
              <appinfo>
                <gml:reversePropertyName>core:connects</gml:reversePropertyName>
              </appinfo>
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NodePropertyType">
    <sequence minOccurs="0">
      <element ref="core:Node"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
```

```
    </complexType>
    <element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type=
  "core:PrimalSpaceLayerType"/>
    <complexType name="PrimalSpaceLayerType">
      <complexContent>
        <extension base="gml:AbstractFeatureType">
          <sequence>
            <element minOccurs="0" name="creationDate" type="dateTime"/>
            <element minOccurs="0" name="terminationDate" type="dateTime"/>
            <element maxOccurs="unbounded" name="cellSpaceMember">
              <annotation>
                <documentation>Cell spaces of a primal space</documentation>
              </annotation>
              <complexType>
                <complexContent>
                  <extension base="gml:AbstractFeatureMemberType">
                    <sequence minOccurs="0">
                      <element ref="core:CellSpace"/>
                    </sequence>
                    <attributeGroup ref="gml:AssociationAttributeGroup"/>
                  </extension>
                </complexContent>
              </complexType>
            </element>
            <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
              <annotation>
                <documentation>Cell boundaries of a primal space</documentation>
              </annotation>
              <complexType>
                <complexContent>
                  <extension base="gml:AbstractFeatureMemberType">
                    <sequence minOccurs="0">
                      <element ref="core:CellBoundary"/>
                    </sequence>
                    <attributeGroup ref="gml:AssociationAttributeGroup"/>
                  </extension>
                </complexContent>
              </complexType>
            </element>
          </sequence>
        </extension>
      </complexContent>
    </complexType>
    <complexType name="PrimalSpaceLayerPropertyType">
      <sequence minOccurs="0">
        <element ref="core:PrimalSpaceLayer"/>
      </sequence>
      <attributeGroup ref="gml:AssociationAttributeGroup"/>
      <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type=
  "core:ThematicLayerType">
      <annotation>
        <documentation>A layer of specific theme aggregating a primal and/or a
  dual space of a given environment.</documentation>
      </annotation>
    </element>
    <complexType name="ThematicLayerType">
      <complexContent>
        <extension base="gml:AbstractFeatureType">
          <sequence>
            <element name="semanticExtension" type="boolean">
              <annotation>
```

```
                <documentation>Indicates whether semantic information is
associated (true) or not (false) to the primal space of the thematic layer.</
documentation>
              </annotation>
            </element>
            <element name="theme" type="core:ThemeLayerValueType">
              <annotation>
                <documentation>Determines the theme of the layer (e.g.,
topographic, logical, etc.).</documentation>
              </annotation>
            </element>
            <element minOccurs="0" name="primalSpace">
              <annotation>
                <documentation>Relates the primal spaces to the thematicLayer</
documentation>
              </annotation>
              <complexType>
                <complexContent>
                  <extension base="gml:AbstractFeatureMemberType">
                    <sequence minOccurs="0">
                      <element ref="core:PrimalSpaceLayer"/>
                    </sequence>
                    <attributeGroup ref="gml:AssociationAttributeGroup"/>
                  </extension>
                </complexContent>
              </complexType>
            </element>
            <element minOccurs="0" name="dualSpace">
              <annotation>
                <documentation>Relates the dual spaces to the thematicLayer.</
documentation>
              </annotation>
              <complexType>
                <complexContent>
                  <extension base="gml:AbstractFeatureMemberType">
                    <sequence minOccurs="0">
                      <element ref="core:DualSpaceLayer"/>
                    </sequence>
                    <attributeGroup ref="gml:AssociationAttributeGroup"/>
                  </extension>
                </complexContent>
              </complexType>
            </element>
          </sequence>
        </extension>
      </complexContent>
    </complexType>
    <complexType name="ThematicLayerPropertyType">
      <sequence minOccurs="0">
        <element ref="core:ThematicLayer"/>
      </sequence>
      <attributeGroup ref="gml:AssociationAttributeGroup"/>
      <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <simpleType name="ThemeLayerValueType">
      <union memberTypes="core:ThemeLayerValueEnumerationType core:
ThemeLayerValueOtherType"/>
    </simpleType>
    <simpleType name="ThemeLayerValueEnumerationType">
      <restriction base="string">
        <enumeration value="Tags"/>
        <enumeration value="Virtual"/>
        <enumeration value="Unknown"/>
```

```
      <enumeration value="Physical"/>
    </restriction>
  </simpleType>
  <simpleType name="ThemeLayerValueOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <simpleType name="TopoExpressionValueType">
    <union memberTypes="core:TopoExpressionValueEnumerationType core:
TopoExpressionValueOtherType"/>
  </simpleType>
  <simpleType name="TopoExpressionValueEnumerationType">
    <restriction base="string">
      <enumeration value="Contains"/>
      <enumeration value="Overlaps"/>
      <enumeration value="Equals"/>
      <enumeration value="Within"/>
      <enumeration value="Crosses"/>
      <enumeration value="Other"/>
    </restriction>
  </simpleType>
  <simpleType name="TopoExpressionValueOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
</schema>
```

Listing B.1

B.2. IndoorGML Indoor Navigation Module

```
<?xml version="1.0" encoding="UTF-8"?>
<schema
    xmlns="http://www.w3.org/2001/XMLSchema"
 xmlns:navi="http://www.opengis.net/indoorgml/2.0/navi"
 xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
 xmlns:gml="http://www.opengis.net/gml/3.2"
 elementFormDefault="qualified"
 targetNamespace="http://www.opengis.net/indoorgml/2.0/navi"
 version="2.0">
  <annotation>
    <appinfo>
      <gml:gmlProfileSchema>IndoorGML</gml:gmlProfileSchema>
    </appinfo>
  </annotation>
  <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://
schemas.opengis.net/iso/19139/20070417/gmd/gmd.xsd"/>
  <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://
schemas.opengis.net/gml/3.2.1/gml.xsd"/>
  <import namespace="http://www.opengis.net/indoorgml/2.0/core" schemaLocation=
"IndoorGML_core.xsd"/>
  <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
  <element name="GeneralSpace" substitutionGroup="navi:NavigableSpace" type=
"navi:GeneralSpaceType"/>
  <complexType name="GeneralSpaceType">
```

```
    <complexContent>
      <extension base="navi:NavigableSpaceType">
        <sequence>
          <element name="function" type="navi:GeneralSpaceFunctionType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="GeneralSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:GeneralSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="GeneralSpaceFunctionType">
    <union memberTypes="navi:GeneralSpaceFunctionEnumerationType navi:
GeneralSpaceFunctionOtherType"/>
  </simpleType>
  <simpleType name="GeneralSpaceFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="Administration"/>
      <enumeration value="Business, trade"/>
      <enumeration value="Education, training"/>
      <enumeration value="Recreation"/>
      <enumeration value="Laboratory"/>
      <enumeration value="Storage"/>
      <enumeration value="Security"/>
    </restriction>
  </simpleType>
  <simpleType name="GeneralSpaceFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <simpleType name="LocomotionAccessType">
    <union memberTypes="navi:LocomotionAccessEnumerationType navi:
LocomotionAccessOtherType"/>
  </simpleType>
  <simpleType name="LocomotionAccessEnumerationType">
    <restriction base="string">
      <enumeration value="Walking"/>
      <enumeration value="Flying"/>
      <enumeration value="Rolling"/>
      <enumeration value="Unspecified"/>
    </restriction>
  </simpleType>
  <simpleType name="LocomotionAccessOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="NavigableBoundary" substitutionGroup="core:CellBoundary" type=
"navi:NavigableBoundaryType"/>
  <complexType name="NavigableBoundaryType">
    <complexContent>
      <extension base="core:CellBoundaryType">
        <sequence>
          <element name="NavigableBoundaryFunction" type="navi:
NavigableBoundaryFunctionType"/>
          <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
        </sequence>
      </extension>
```

```
    </complexContent>
  </complexType>
  <complexType name="NavigableBoundaryPropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NavigableBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="NavigableBoundaryFunctionType">
    <union memberTypes="navi:NavigableBoundaryFunctionEnumerationType navi:Naviga
bleBoundaryFunctionOtherType"/>
  </simpleType>
  <simpleType name="NavigableBoundaryFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="AnchorBoundary"/>
      <enumeration value="ConnectionBoundary"/>
    </restriction>
  </simpleType>
  <simpleType name="NavigableBoundaryFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="NavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NavigableSpaceType"/>
  <complexType name="NavigableSpaceType">
    <complexContent>
      <extension base="core:CellSpaceType">
        <sequence>
          <element name="locomotionType" type="navi:LocomotionAccessType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NavigableSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NavigableSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="NonNavigableBoundary" substitutionGroup="core:CellBoundary"
type="navi:NonNavigableBoundaryType"/>
  <complexType name="NonNavigableBoundaryType">
    <complexContent>
      <extension base="core:CellBoundaryType">
        <sequence/>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NonNavigableBoundaryPropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NonNavigableBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="NonNavigableSpace" substitutionGroup="core:CellSpace" type=
"navi:NonNavigableSpaceType"/>
  <complexType name="NonNavigableSpaceType">
    <complexContent>
      <extension base="core:CellSpaceType">
```

```
        <sequence/>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NonNavigableSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NonNavigableSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="ObjectSpace" substitutionGroup="navi:NonNavigableSpace" type=
"navi:ObjectSpaceType"/>
  <complexType name="ObjectSpaceType">
    <complexContent>
      <extension base="navi:NonNavigableSpaceType">
        <sequence>
          <element minOccurs="0" name="description" type="string"/>
          <element minOccurs="0" name="containedFeatures" type="integer"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="ObjectSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:ObjectSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Route" substitutionGroup="gml:AbstractFeature" type="navi:
RouteType"/>
  <complexType name="RouteType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="creationDate" type="dateTime"/>
          <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type=
"core:NodePropertyType"/>
          <element maxOccurs="unbounded" name="routeEdge" type="core:
EdgePropertyType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="RoutePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:Route"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="TransferSpaceType">
    <union memberTypes="navi:TransferSpaceEnumerationType navi:
TransferSpaceOtherType"/>
  </simpleType>
  <simpleType name="TransferSpaceEnumerationType">
    <restriction base="string">
      <enumeration value="Door"/>
      <enumeration value="Window"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceOtherType">
```

```xml
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="TransferSpace" substitutionGroup="navi:NavigableSpace" type=
"navi:TransferSpaceType"/>
  <complexType name="TransferSpaceType">
    <complexContent>
      <extension base="navi:NavigableSpaceType">
        <sequence>
          <element name="function" type="navi:TransferSpaceFunctionType"/>
          <element name="type" type="navi:TransferSpaceType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="TransferSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:TransferSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="TransferSpaceFunctionType">
    <union memberTypes="navi:TransferSpaceFunctionEnumerationType navi:TransferSp
aceFunctionOtherType"/>
  </simpleType>
  <simpleType name="TransferSpaceFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="ConnectionSpace"/>
      <enumeration value="AnchorSpace"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
</schema>
```

Listing B.2

C

ANNEX C (INFORMATIVE) EXAMPLE OF CODELIST

ANNEX C (INFORMATIVE) EXAMPLE OF CODELIST

For mapping indoor space features to IndoorNavigation module classes, space features have to be classified by functions and usages of space. Because misspellings or different names for the same notion brings the problems in data interoperability, in order to overcome the typo errors, an example of CodeList is provided in this annex to specify the attribute values including space category types and space function types. The CodeLists are defined from OmniClass (Table 13 and Table 14) created and used by the North American architectural, engineering and construction (AEC) industry.

C.1. GeneralSpace

C.1.1. GeneralSpaceFunctionType

Code list derived from OmniClass Table 13:

- T1000 — Corridor
- T1010 — Breezeway
- T1020 — Box Lobby
- T1030 — Elevator Lobby
- T1040 — Landing
- T1050 — Aisle
- T1060 — Ramp
- T1070 — Concourse
- T1080 — Moving walkway
- T1090 — Entry Lobby
- T1100 — Jet way
- T1110 — Elevator Shaft

- T1120 — Stair
- T1130 — Chute
- G1000 — Elevator Machine Room
- G1010 — Fire Command Center
- G1020 — Men’s Restroom
- G1030 — Unisex Restroom
- G1040 — Refrigerant Machinery Room
- G1050 — Incinerator Room
- G1060 — Gas Room
- G1070 — Liquid Use, Dispensing and Mixing Room
- G1080 — Electrical Room
- G1090 — Telecommunications Room
- G1100 — Hazardous Waste Storage
- G1110 — Building Manager Office
- G1120 — Guard Stations
- G1130 — Women’s Restroom
- G1140 — Furnace Room
- G1150 — Fuel Room
- G1160 — Liquid Storage Room
- G1170 — Hydrogen Cutoff Room
- G1180 — Switch Room
- G1190 — Classrooms
- G1200 — Assembly Hall
- G1210 — Physics Teaching Laboratory
- G1220 — Open Class Laboratory
- G1230 — Laboratory Service Space
- G1240 — Computer Lab
- G1250 — Training Support Space

- G1260 — Study Room
- G1270 — Evidence Room
- G1280 — Witness Stand
- G1290 — Robing Area
- G1300 — Council Chambers
- G1310 — Armory
- G1320 — General Performance Spaces
- G1330 — Orchestra Pit
- G1340 — Performance Hall
- G1350 — Audience Space
- G1360 — Audience Seating Space
- G1370 — Projection Booth
- G1380 — Stage Wings
- G1390 — Art Gallery
- G1400 — Sculpture Garden
- G1410 — Recording Studio
- G1420 — Photo Lab
- G1430 — Library
- G1440 — Mediation Chapel
- G1450 — Reflection Space
- G1460 — Chapel
- G1470 — Shrine
- G1480 — Confessional Space
- G1490 — Tabernacle
- G1500 — Choir Loft
- G1510 — Marriage Sanctuary
- G1520 — Mental Health Quiet Room
- G1530 — Bone Densitometry Room

- G1540 — CT Simulator Room
- G1550 — Head Radiographic Room
- G1560 — Mobile Imaging System Alcove
- G1570 — MRI System Component Room
- G1580 — PET/CT Simulator Room
- G1590 — Radiographic Room
- G1600 — Stereotactic Mammography Room
- G1610 — Ultrasound/Optical Coherence Tomography Room
- G1620 — Angiographic Control Room
- G1630 — Angiographic Procedure Control Area
- G1640 — Silver Collection Area
- G1650 — Computer Image Processing Area
- G1660 — CT Control Area
- G1670 — Image Quality Control Room
- G1680 — X-Ray, Plane Film Storage Space
- G1690 — MRI Control Room
- G1700 — MRI Viewing Room
- G1710 — Radiographic Control Room
- G1720 — Tele-Radiology/Tele-Medicine Room
- G1730 — Radiation Diagnostic and Therapy Spaces
- G1740 — Health Physics Laboratory
- G1750 — Linear Accelerator Entrance Maze, Healthcare
- G1760 — Radioactive Waste Storage Room, Healthcare
- G1770 — Nuclear Medicine Scanning Room
- G1780 — Patient Dose/Thyroid Uptake Room
- G1790 — Radiopharmacy
- G1800 — Radiation Therapy, Mold Fabrication Shop
- G1810 — Hearth and Lung Diagnostic and Treatment Spaces

- G1820 — Cardiac Catheter Instrument Room
- G1830 — Cardiac Catheter Control Room
- G1840 — Cardiac Electrophysiology Room
- G1850 — Echocardiograph Room
- G1860 — Extended Pulmonary Function Testing Laboratory
- G1870 — Pacemaker ICD Interrogation Room
- G1880 — Procedure Viewing Area
- G1890 — Pulmonary Function Treadmill Room
- G1900 — Respiratory Therapy Clean-up Room
- G1910 — General Diagnostic Procedure and Treatment Spaces
- G1920 — Endoscopy/Gastroenterology Spaces
- G1930 — Clinical Laboratory Spaces
- G1940 — Pharmacy Spaces
- G1950 — Rehabilitation Spaces
- G1960 — Medical Research and Development Spaces
- G1970 — Chemistry Laboratories
- G1980 — Physical Sciences Laboratories
- G1990 — Earth and Environmental Sciences Laboratories
- G2000 — Psychology Laboratories
- G2010 — Dry Laboratories
- G2020 — Wet Laboratories
- G2030 — Biosciences Laboratories
- G2040 — Astronomy Laboratories
- G2050 — Forensics Laboratories
- G2060 — Bench Laboratories
- … **(Refer to the full standard implementation schema for other classifications)**

C.2. TransferSpace

C.2.1. TransferSpaceCategoryType

- C1000 — Door
- C1010 — Vestibule
- C1020 — Sally port
- A1000 — Vestibule
- A1020 — Gate

C.2.2. TransferSpaceFunctionType

- C1000 — Door
- C1010 — Vestibule
- C1060 — Sally port
- A1000 — Entry Vestibule
- A1010 — Exterior door
- A1020 — Gate
- A1030 — Emergency door

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BIBLIOGRAPHY

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